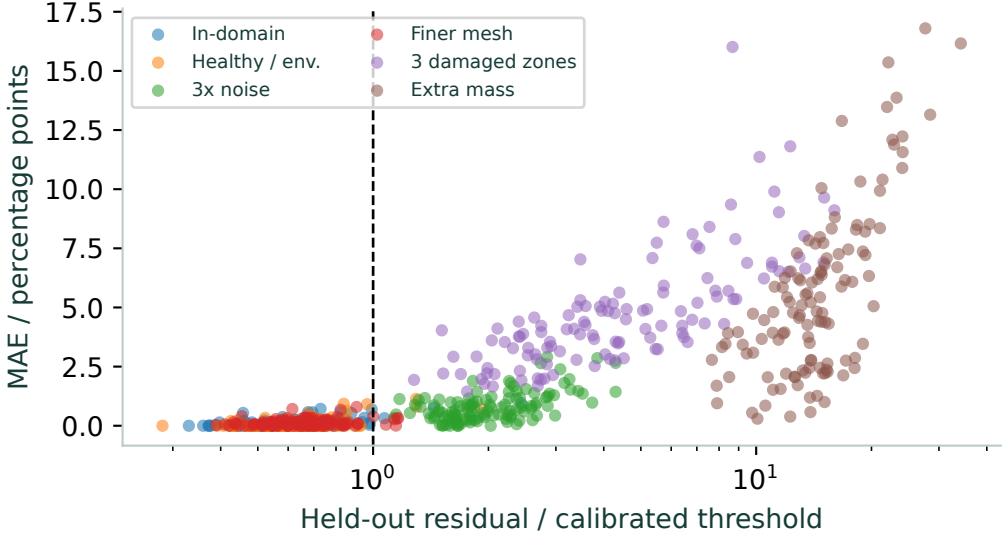
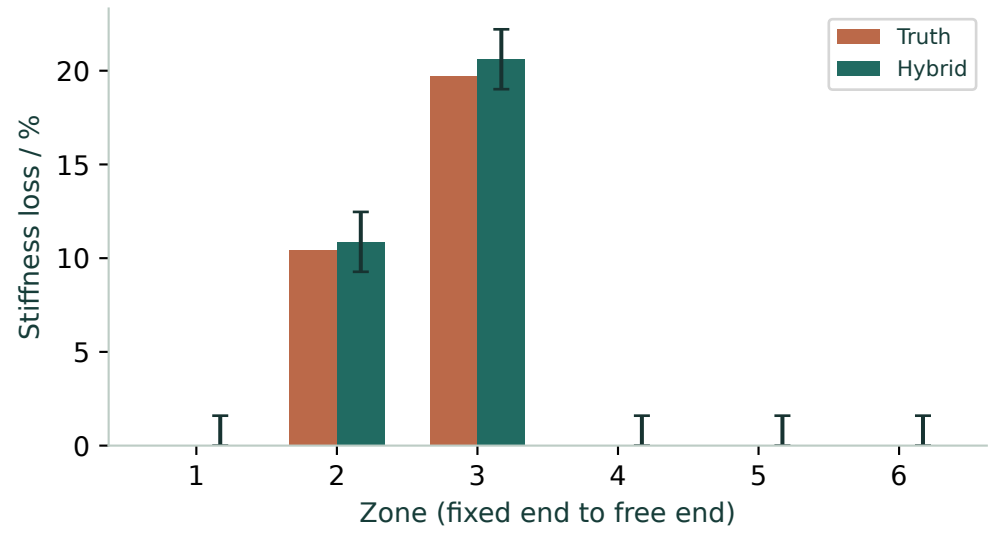

In-domain
Healthy / env.
3x noise
Finer mesh
3 damaged zones
Extra mass
MAE / percentage points
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10^0
10^1
Held-out residual / calibrated threshold


Predefined example: first two-damage case, seed 11
Truth
Hybrid
Stiffness loss / %
20
15
10
5
0
1
2
3
4
5
6
Zone (fixed end to free end)